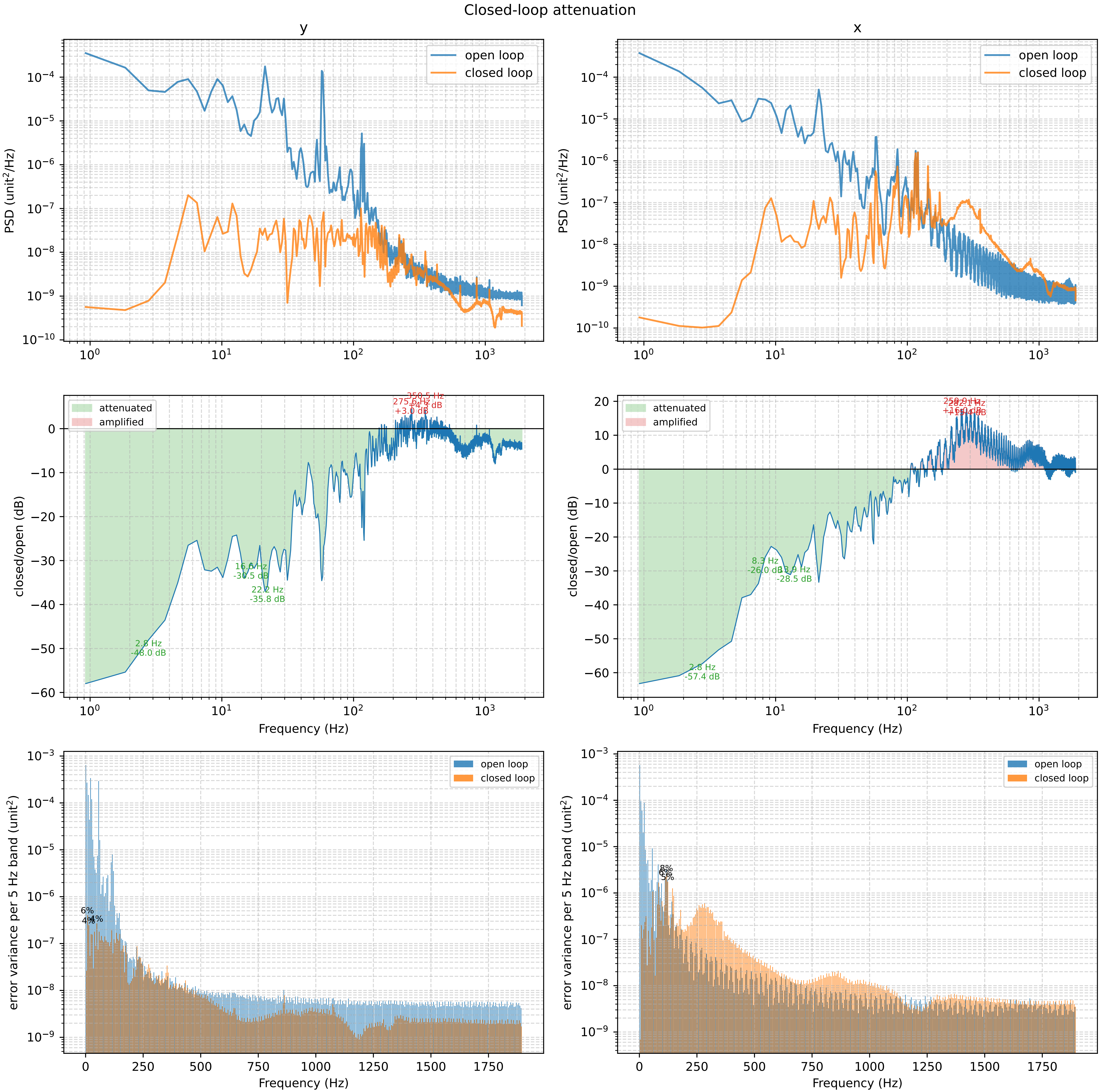

Closed-loop attenuation

Run configuration and per-band RMS

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Started: 2026-07-31 13:39:19
pixel_scale: 15.5 px per output unit
y: mean +3.486 -> -0.000 px;  RMS about mean 2.513 -> 0.039 px   (open -> closed)
x: mean +10.914 -> -0.000 px;  RMS about mean 6.700 -> 0.091 px   (open -> closed)

Config used:
ATTENUATION PAR-5 -- PAR-4 REPEATED with the beam RE-CENTRED (start_y 684 -> 692, park verified +0.70 px y / +0.53 px x, against PAR-4's +5.95 px which ran the 20x20 tracking window off the ROI edge and halved y's rejection in every band) and with the NLMS step SCHEDULED 0.5 -> 0.03 over 60 s to fix the convergence PAR-4's trace showed was still incomplete. Kx stays 4.05 -- PAR-4 proved it: x improved in every band (60-150 Hz 1.4x -> 2.9x, 150-250 Hz from 0.4x amplifying to 1.3x rejecting) and beat its environment-scaled predicted asymptote by 11%. Timeline unchanged from PAR-4 so the scored windows are directly comparable. Through the PARPORT DAC (TI DAC81404 over the StarTech PEX1P2 / AX99100 PCIe parallel card, spi link, mmio backend), replacing the DAQC2/BRIDGEplate serial path used by every previous attenuation run.
TRANSPORT: bit-banged 24-bit SPI, ~6.4 us per channel update (~12.8 us for both channels), against the BRIDGEplate's ~0.5-1.4 ms. DAC range is FIXED at +-5 V bipolar (153 uV/LSB) with the mirror parked at 0 V, matching the bode sweeps exactly; auto-ranging is off, so the actuator is a fixed-gain element for both phases of the A/B and the open/closed comparison is clean. Supplies +/-12 V: AVDD needs VMAX+1.5 = 11.5 V for +-10, AVSS -11.5 V, leaving only 0.5 V of margin -- meter the rails under load. AVSS is therefore NEGATIVE, not grounded, which means the 0-5 base range runs outside the AVSS = 0 V condition SLASEH2A 7.5 gives the unipolar figures at (see _avss_tradeoff); 16-bit = 76 uV/LSB in 0-5, 305 uV/LSB in +-10, vs the BRIDGEplate's 12-bit 1 mV.
OUTPUTS: two channels, no coarse stage -- ch0 = y/tip (+1.0 V/unit), ch1 = x/tilt (-1.0 V/unit), offset 0 V, symmetric command bounds +-1 V (59 px of authority against a +-16 px ROI; the mirror's full range is deliberately NOT given to the loop on this first run). NOTE the command->FSM voltage mapping is NOT the same as any previous attenuation run, so K is not comparable across the transport change and was re-measured -- and it came out 4.7-5.5x HIGHER, not lower as this file originally predicted, so nothing about the piplate-era gains carries over.
Camera/CoM: 32x32 ROI start_x 1816 / start_y 684, exposure 50, gain 100, bw 100, 20x20 tracked window thr 5 (the camera is now shrouded, so the background floor no longer needs a high threshold). Identical to conf_bode_par_x/y[_small].json, so K and tau were fit through exactly this optical chain.
FSP: broad_order 512, broad_mu .03 -- REVERTED from PAR-2's 128/.005, which was 60% worse on y and 10% on x once normalised for the environment. 128 taps = 29.6 Hz resolution cannot separate the 2-3 Hz-spaced 9-30 Hz comb; this reproduces the documented kalman order-80 failure (order 80 ~29 Hz could not represent the 2.256 Hz comb). broad_lp 5 (retuned: the shorter delay moved the regeneration band from ~325 Hz to ~700 Hz); adapt_period 8, adapt_tau 5. Guard ON (guard_ratio 4). Cold-start observer: pass_open false + broad_freeze_closed false, so no weight is derived from open-loop data; drift_tau and the transient path are OFF. Modal freqs re-derived from run 1's own open PSD (8-mode cap), though they are inert while broad_order > 0.
PURPOSE: isolate the two things PAR-4 confounded. PAR-4 changed Kx AND happened to run with the beam 5.95 px off centre; x improved everywhere (the Kx fix) while y lost half its rejection uniformly across 1-60 Hz (the clipped tracking window). PAR-5 fixes the centring and changes nothing else about the controller except the step schedule, which the offline replay shows does not move the converged solution. EXPECTATIONS: y should return to a closed/open ratio near PAR-3's 0.121 (PAR-4 was 0.233); x should hold or beat PAR-4's 0.248 (PAR-3 was 0.302). Prior absolute numbers are NOT directly comparable because the environment moves run to run -- PAR-4's open was 18% (y) and 34% (x) larger than PAR-3's, and a new 150-250 Hz component appeared that carried 8.7x (y) and 5x (x) more open power than in PAR-3. Always normalise the closed RMS by that run's own open RMS before claiming an improvement.
DELAY AND K: taken from different sweeps on purpose. DELAY from the newest full 5-300 Hz sweeps, fsm_bodex_par2 / fsm_bodey_par2 (2026-07-31 12:23 / 12:16): x 2.763 fr (tau 0.4654 ms), y 2.624 fr (tau 0.4288 ms) at fs 3788. PAR-1..3 ran 2.717 / 2.577; the +0.046 fr shift sits inside the fits' own phase residuals, so the two sweep pairs agree and the newer/longer value is kept only because under-modeled delay is the documented failure mode. K from the SMALL-SIGNAL plateau: Kx 4.05, Ky 3.646. That distinction is not academic -- the loop commands ~6 mV rms / ~20 mV peak and the FSM compresses ~4% at +-100 mV, but WHICH AXIS compresses moves between days: on 07-30 it was x (100/50 mV ratio 1.041, y flat at 0.998), on 07-31 it was y (ratio 1.039, x flat at 1.003). The raw 07-31 fit lines therefore read Kx 4.050 (correct) and Ky 3.510 (3.7% low), and PAR-4 uses 3.646 for y from the +-20 mV sweep, corroborated by the same full file's +-50 mV pass at 3.662. Conversely tau is NOT measurable from a small sweep -- 10-30 Hz spans ~7 deg of phase and their loop_frames scatters 1.09-2.02 fr -- which is why delay comes only from the full sweeps. Against the piplate era this is ~5x MORE loop gain and LESS THAN HALF the delay; the shorter delay is the entire reason to expect better rejection here. Caveat: the small-signal gain itself wanders a few percent per hour (seven +-20 mV x sweeps read 3.97-4.36, each internally consistent), and erring HIGH on K is the safe direction since u = -phi_hat/K.
COMPARE against attenuation_fsp21 and attenuation_fsp15 (run 15: y 0.073 / x 0.168 px): report total and per-band closed RMS, 60-400 Hz waterbed, command/clamp behavior, and transient activation count on the shared 500/500/4096 grid. Closed RMS in PIXELS remains comparable across the change; command units do not. refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010
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RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.4675	0.0102	0.4009	0.0070
10-20	0.2148	0.0085	0.1393	0.0080
20-30	0.3314	0.0087	0.1525	0.0115
30-40	0.0753	0.0071	0.0476	0.0066
40-50	0.0411	0.0093	0.0256	0.0053
50-60	0.2687	0.0078	0.0514	0.0174
60-70	0.0645	0.0079	0.0177	0.0052
70-80	0.0327	0.0074	0.0260	0.0127
80-90	0.0230	0.0079	0.0364	0.0232
90-100	0.0358	0.0072	0.0239	0.0149
100-110	0.0175	0.0068	0.0147	0.0133
110-120	0.0566	0.0078	0.0356	0.0344
120-130	0.0316	0.0074	0.0228	0.0223
130-140	0.0129	0.0084	0.0106	0.0106
140-150	0.0138	0.0081	0.0155	0.0225
150-160	0.0089	0.0059	0.0078	0.0090
160-170	0.0067	0.0052	0.0088	0.0100
170-180	0.0072	0.0061	0.0067	0.0130
180-190	0.0049	0.0026	0.0080	0.0088
190-200	0.0048	0.0027	0.0043	0.0086
200-210	0.0048	0.0034	0.0072	0.0093
210-220	0.0043	0.0034	0.0036	0.0096
220-230	0.0055	0.0057	0.0063	0.0103
230-240	0.0050	0.0049	0.0040	0.0120
240-250	0.0043	0.0042	0.0055	0.0139
250-260	0.0034	0.0031	0.0041	0.0152
260-270	0.0034	0.0035	0.0044	0.0152
270-280	0.0032	0.0040	0.0045	0.0159
280-290	0.0033	0.0031	0.0037	0.0160
290-300	0.0031	0.0027	0.0045	0.0155
total 0-300	0.6829	0.0352	0.4634	0.0799